

CONSTRUCTION SCOPE OF WORK PROJECT 537-26-702, CHICAGO, IL. EHRM INSTALLATION OF DATA DROP AND POWER OUTLETS (DDPO)

1. GENERAL INTENT:

The Contractor shall furnish all labor, materials and equipment to prepare the site identified on the Jesse Brown VA campus for the installation of additional data drops, power outlets and equipment bracing for equipment deployment of the Electronic Health Record (EHR) Management System.

2. STATEMENT OF WORK:

Provide and install normal and emergency power outlets, data drops, and required equipment bracing to support EHR equipment deployment on the Jesse Brown VA campus as shown and described in the project drawings and specifications. A list of data drops and power outlets will be provided within the solicitation documents.

PLEASE NOTE: The contractor is responsible for damage caused by his actions to the walls, floors, ceilings, and roofs. The contractor whose work causes damage is responsible for patching to match original construction, fire rating, and finish.

Where existing electrical systems are in areas that conflict with new equipment, piping, or ductwork to be installed, the contractor shall either arrange new equipment, conduit, or ductwork in such a fashion that it does not conflict with existing systems, or rework existing electrical systems to allow for installation of new equipment, piping, or ductwork.

Additionally, the contractor shall obtain approval from the COR before turning off the power to equipment, systems, panels, etc. affected by the installation of new devices. Thus, the contractor shall coordinate all outages with the COR.

The contractor shall verify accessibility to the area of his/her work and shall notify the COR prior to if other utilities are required to be removed or relocated to allow access to his/her area of work.

Where existing electrical systems are in areas that conflict with new equipment, piping, or ductwork to be installed, the contractor shall either arrange new equipment, conduit, or ductwork in such a fashion that it does not conflict with existing systems, or rework existing electrical systems to allow for installation of new equipment, piping, or ductwork.

Contractor shall be responsible for all openings required in walls. All openings shall be repaired to match existing by a qualified contractor at the expense of this contractor. All conduits through walls shall be grouted or sealed into openings.

Devices within or adjacent to the path of construction shall be protected in place. Devices that must be removed shall be tested prior to removal, protected from damage, and re-installed in its original location.

Existing conditions are shown based on information obtained from field surveys, existing building documents, and staff. Not all existing equipment and conduit are shown. **Verify existing conditions and report any conflicts before proceeding.**

Contractor shall verify all furniture, modular furniture, and equipment locations with architectural plans, elevations, and reviewed shop drawings. Prior to making the actual electrical installation, the contractor shall adjust receptacles, outlets, or connection locations to accommodate furniture and/or equipment.

Flush mount all receptacles at the existing height of other receptacles in the room, except where otherwise noted. Receptacles and outlets may be surface mounted when conduit is specified exposed.

Electrical and technology equipment shall be mounted to avoid impedance of, operation of, and/or access to electrical and mechanical equipment. All mounting of electrical and telecommunications equipment, on equipment supplied by another contractor, shall be approved in advance by the other contractor.

Flush mount all telecommunication outlets at +18" from floor (centerline dimension). Outlets may be surface mounted when conduit is specified exposed where required when telecommunication outlets are to be mounted in concrete walls.

Coordinate with COR for installation work performed in a high-risk area. Additionally, work that shall be completed after hours and on weekends must be approved by COR. **Normal working hours are 7:30am to 4:00pm CST, Monday-Friday.**

Contractor shall follow Infection Control procedures as outlined in the VA PRA/ICRA provided prior to construction.

2.1 Data Drops

- 2.1.1 Install Cat 6A cabling and data drops in accordance with attachments, location drawings, and specifications.
- 2.1.2 Horizontal cabling shall be Category 6A (Cat6A) (ANSI/TIA-1179 Healthcare Facility Telecommunications Infrastructure).
- 2.1.3 Installation shall be compliant with the BICSI standards (ANSI/NECA/BICSI 568, Standard for Installing Commercial Building Telecommunications Cabling and ANSI/TIA-1179-A Healthcare Facility Telecommunications Infrastructure for new installations).
 - 2.1.3.1 New cable installations additionally shall be compliant with TIA 568.2-D, TIA TSB-184-A, and NFPA-70 Section 725.144 to prevent PoE heat accumulation. Provide certification reports to the COR.
 - 2.1.3.2 Alternatively, new UTP cabling shall carry a UL listed "LP" (Limited Power) rating.

- 2.1.3.3 Cat6A cable shall be terminated to identified RJ-45 patch panel in IT closet and CAT6A RJ-45 jack at wall plate and labeled at both ends.
- 2.1.3.4 Use a cable analyzer to determine cable length (max cable length 90M/295 linear feet) and certify that cable meets or exceeds TIA-EIA-568-C.2.
- 2.1.3.5 Cables shall be installed in continuous lengths from origin to destination, no splices with a 1' service loop.
- 2.1.5 Cable management shall be implemented for all patch cables in accordance with [*Infrastructure Standard for Telecommunications Spaces*](#) (latest version on the TIL), ensuring proper bend radius, bundling, airflow, support and labeling.
- 2.1.6 Copper structured cabling shall be certified as meeting or exceeding TIA-EIA-568.1-D (max fixed cable length 90M/295 linear feet) independent of media type.
- 2.1.7 Use existing raceways as much as possible. Cable shall be installed in accordance with manufacturer's recommendations and best industry practices. Existing raceways shall not be filled greater than the NEC maximum fill for the raceway type.
- 2.1.8 Install and label outlets, faceplates and patch panels in accordance with specifications and VAMC standard. Patch panel labels shall include room #, outlet #, jack #. The Contractor shall submit a labeling plan for outlets, faceplates and patch panels to the COR for approval prior to initiating installation. All label printing shall be machine generated.
- 2.1.9 Confirm locations, including installation height, with COR.
- 2.1.10 Patch and finish disturbed surfaces to match existing.
- 2.1.11 All new cable installation shall be tested and certified. Provide the certification report to COR.
- 2.1.12 All materials used to seal penetrations of fire rated walls and floors shall be tested and certified as a system per ASTM E814 Standards for Fire Tests of Through-Penetration Firestops.
- 2.2 Power Outlets
 - 2.2.1 Install conduit and conductors in accordance with attachments, location drawings, and specifications.
 - 2.2.2 Use existing raceways as much as possible. Conductors shall be installed in accordance with manufacturer's recommendations and best industry practices. Existing raceways shall not be filled greater than the NEC maximum fill for the raceway type.

- 2.2.3 Install and label outlets, faceplates and power panels in accordance with specifications and VAMC standard. The Contractor shall submit a labeling plan for outlets, faceplates and power panels to the COR for approval prior to initiating installation. All label printing shall be machine generated.
- 2.2.4 Confirm locations, including installation height, with COR.
- 2.2.5 Patch and finish disturbed surfaces to match existing.
- 2.2.6 All materials used to seal penetrations of fire rated walls and floors shall be tested and certified as a system per ASTM E814 Standards for Fire Tests of Through-Penetration Firestops.
- 2.2.7 Life safety, critical, equipment branch wiring for feeders and branch circuits shall be routed in separate raceway, junction boxes, pull boxes, and cabinets. Wiring for each branch shall be independent from other branches, including the normal branch.

2.3 Equipment and Mounting Hardware

- 2.3.1 List of equipment and mounting hardware requiring installation is located within the solicitation attachments. **Equipment listed is Government-Furnished; mounting hardware for equipment shall be provided by Contractor unless otherwise noted by COR. Locations are shown on the solicitation attachments, but shall be verified with COR prior to installation.**
- 2.3.2 Install units in accordance with manufacturer's installation instructions.
- 2.3.3 Patch and finish disturbed surfaces to match existing.

3. PERIOD OF PERFORMANCE:

- 3.1 Work timelines and completion progress shall be communicated to the COR for tracking purposes.
- 3.2 Work described in Section 2 shall be completed within 60 days following Notice to Proceed.

4. REFERENCES:

- 4.1 VA OEHRM [*Site Infrastructure and End User Device \(EUD\) Requirements*](#)
- 4.2 [*Infrastructure Standards for Telecommunications Spaces*](#)
- 4.3 *Telecommunications and Special Telecommunications Systems Design Manual*, <https://www.cfm.va.gov/til/dManual/dmTelecomm.pdf>.

5. SUBMITTALS:

- 5.1 In addition to submittals required by specifications, the contractor shall submit the following submittals for review:
 - 5.1.1 Product data and sample of cable.
 - 5.1.2 As-built drawings for all data locations and labels. The Contractor shall provide the central drawing set to the Owner at the end of the project. The marked-up drawing set will accurately depict the as-built status of the system including termination locations, cable routing, and all administration labeling for the cabling system. In addition, a narrative will be provided that describes any areas of difficulty encountered during the installation that could potentially cause problems to the telecommunications system.
 - 5.1.3 As-built drawings for new power locations. Include one line diagram depicting which circuits feed new outlets and panel schedules for panels feeding new outlet locations.
 - 5.1.4 Certification report. All testing documents shall reflect the appropriate labeling scheme.
 - 5.1.5 Warranties of installation and cabling system. The Contractor shall provide a system warranty covering the installed cabling system against defects in workmanship, components, and performance, and covering follow-on support after project completion.

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